

Clone tDRMapper from GitHub repository:
<https://github.com/sararselitsky/tDRmapper> to local

conda activate bioinfo #activate virtual environment

cd your_sample_directory

```
perl /mnt/f/SARS-COV2\sequencing/tDRmapper/Scripts/TdrMappingScripts.pl  
/mnt/f/SARS-COV2\sequencing/tDRmapper/mm10_mature_pre_for_tdrMapper.fa  
your_sample.fastqsanger.gz
```

```
perl /mnt/f/SARS-COV2\sequencing/tDRmapper/Scripts/TdrMappingScripts.pl  
/mnt/f/SARS-COV2\sequencing/tDRmapper/mm10_mature_pre_for_tdrMapper.fa  
your_sample.fastqsanger.gz
```

... run each or loop for all samples

#Rename 1-12 based on treatment

```
echo -e  
"tRNA_ID\tC1\tC2\tC3\tC4\tC5\tyour_treatment_prefix1\tyour_treatment_prefix2\tyour_treatment  
_prefix3\tyour_treatment_prefix4\tyour_treatment_prefix5" > tRNA_count_matrix.txt
```

```
awk 'BEGIN {  
    FS = OFS = "\t";  
    split("C1 C2 C3 C4 C5 your_treatment_prefix1 your_treatment_prefix2  
your_treatment_prefix3 your_treatment_prefix4 your_treatment_prefix5", samples, " ");  
}  
FNR == 1 { next } # Skip header  
{  
    tRNA[$2]; # Collect all tRNA IDs  
    sample = (FILENAME ~ /S1_F1/) ? "C1" : #replace with control prefixes  
        (FILENAME ~ /S2_F1/) ? "C2" :  
        (FILENAME ~ /S3_F1/) ? "C3" :  
        (FILENAME ~ /S4_F1/) ? "C4" :  
        (FILENAME ~ /S5_F1/) ? "C5" :  
        (FILENAME ~ /your_treatment_sample1/) ? "your_treatment_prefix1" :  
        (FILENAME ~ /your_treatment_sample2/) ? "your_treatment_prefix2" :  
        (FILENAME ~ /your_treatment_sample3/) ? "your_treatment_prefix3" :  
        (FILENAME ~ /your_treatment_sample4/) ? "your_treatment_prefix4" :  
        (FILENAME ~ /your_treatment_sample5/) ? "your_treatment_prefix5" : "UNKNOWN";  
    if (sample == "UNKNOWN") next;
```

```

    counts[$2][sample] += $4;
}
END {
    for (id in tRNA) {
        printf "%s", id;
        for (i = 1; i <= length(samples); i++) {
            s = samples[i];
            printf "\t%d", (s in counts[id]) ? counts[id][s] : 0;
        }
        printf "\n";
    }
}' \
C1.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
C2.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
C3.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
C4.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
C5.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
your_treatment_sample1.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
your_treatment_sample2.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
your_treatment_sample3.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
your_treatment_sample4.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt \
your_treatment_sample5.fastqsanger.gz.hq_cs.mapped.speciesInfo.txt >>
tRNA_count_matrix.txt

```

Now continue to DE analysis using tDRMapper_DE.R script.